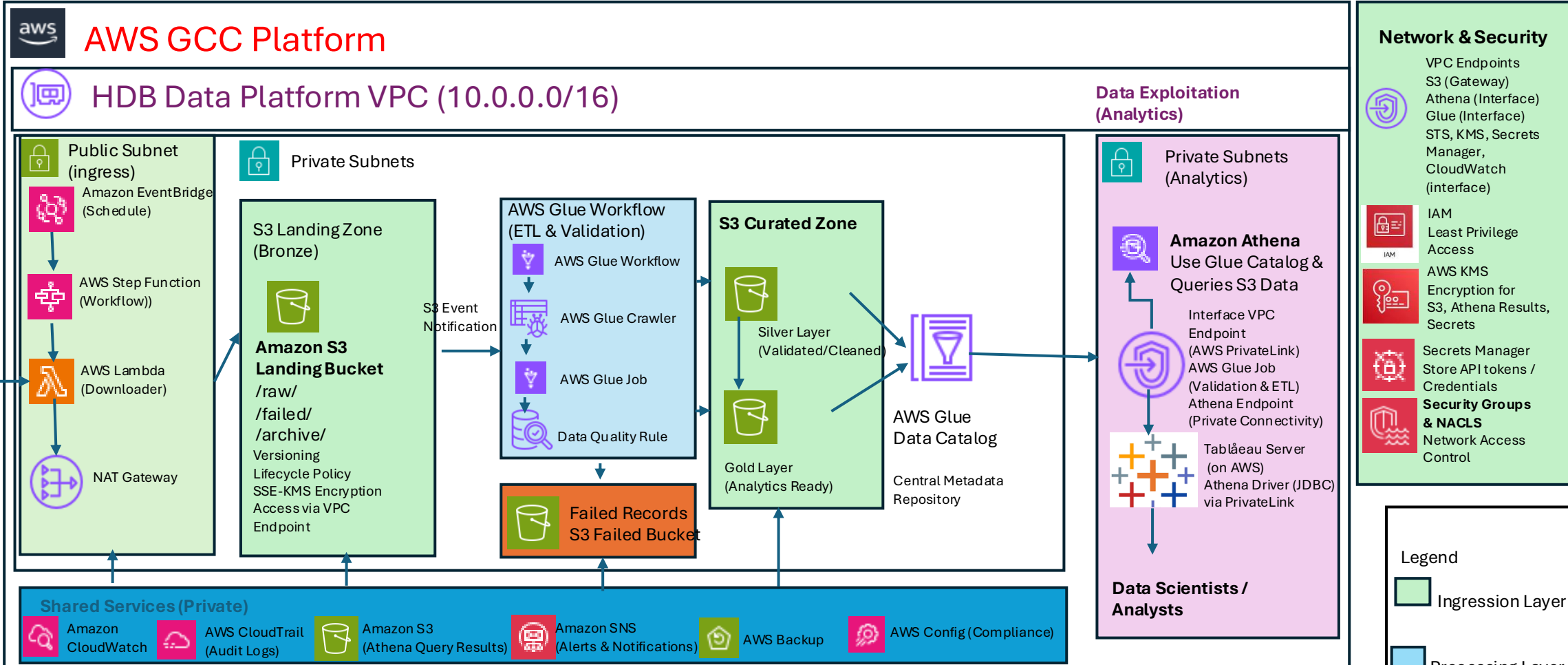


HDB Data Platform - Data Ingestion & Data Exploitation Architecture on AWS

Secure • Scalable • Performant • Operational Excellence



Data Flow Summary

- 1.EventBridge triggers Step Functions on schedule
- 2.Lambda downloads data from data.gov.sg (HTTPS) using streaming for large files and uploads to S3 Landing
- 3.S3 event triggers Glue Workflow
- 4.Data is validated and transformed by Glue Jobs
- 5.Clean data stored in Silver(validated) and Gold (analytics)
- 6.Glue Catalog stores metadata
- 7.Athena queries data in S3 using Glue Catalog
- 8.Tableau connects to Athena via PrivateLink for analysis

Security & Governance

 Encryption At Rest (S3 SSE-KMS) In Transit (TLS 1.2+)	 Least Privilege IAM Roles for each service	 Data Governance Glue Catalog, Data Quality Rules	 Logging & Auditing CloudTrail, CloudWatch, AWS Config
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Key Design Principles

All processing and analytics components are in private subnets

No inbound access from internet

VPC Endpoints ensure private communication with AWS services

S3 encryption, IAM least privilege, logging and monitoring enabled Scalable, serverless where possible (Lambda, Glue, Athena)

Cost optimized with S3 lifecycle, partitioning and serverless services